

$$\beta_n = \beta_p = KP_n \frac{W_1}{L_1} = KP_p \frac{W_2}{L_2}$$

Since $KP_n = 3KP_p$, the width of the PMOS device must be three times the width of the NMOS, assuming equal-length MOSFETs. For $V_{SP} = 2.5$ V, this requires

$$W_2 = 3W_1$$

which is also the requirement for making $R_n = R_p$. This is an important practical result. It shows how the electron and hole mobilities are related to both the effective switching resistances and the switching point voltage. As seen in Table 10.2, we often increase the widths of the PMOS devices to try to center V_{SP} and equate the propagation delays (the pull-up resistance, R_p , is the same as the pull-down resistance R_n). ■

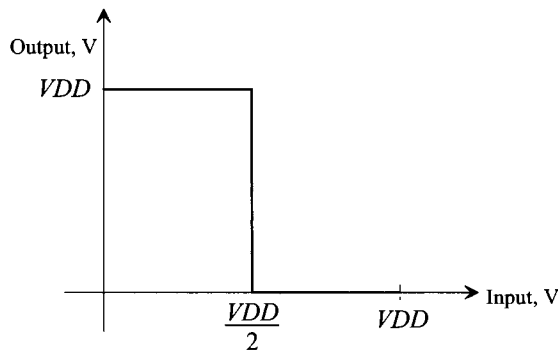


Figure 11.6 Ideal VTCs for the inverter.

Example 11.3

Show, using SPICE, that the inverter seen in Fig. 11.7 and implemented in the short-channel CMOS process has a switching point voltage close to the ideal value of $V_{DD}/2$. Comment on the sizes of the devices.

The simulation results are seen (also) in Fig. 11.7. Because we've sized the width of the PMOS to twice the width of the NMOS (see Table 10.2), the switching point is close to the ideal value of $V_{DD}/2$. We know that short-channel devices don't follow the square-law models and so we can't use Eq. [11.4] to calculate the V_{SP} in our 50 nm process. We can, however, get a good estimate using the effective switching resistances as seen in the figure (a voltage divider). When $R_p = R_n$, the switching point voltage is close to ideal. By changing the widths of the devices, we can adjust the switching point voltage. In other words, for a short-channel CMOS process we can use

$$V_{SP} = V_{DD} \cdot \frac{R_n}{R_n + R_p} \quad (11.5)$$

to estimate V_{SP} . This equation does have limitations. For example, if $R_n \gg R_p$, then this equation indicates V_{SP} is V_{DD} . However, we know that V_{SP} has to be

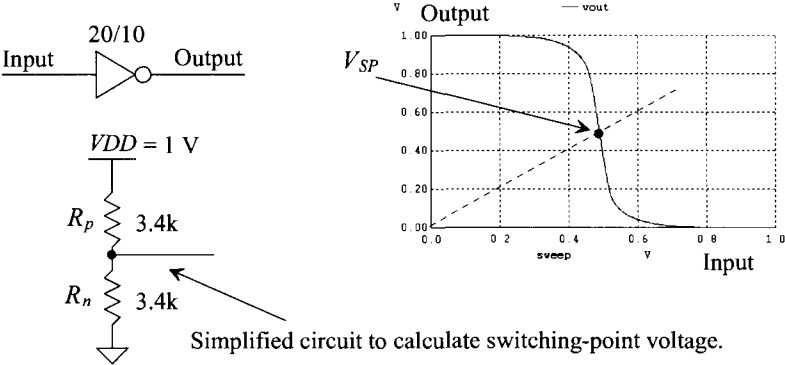


Figure 11.7 Switching point voltage for an inverter in the short-channel process.

greater than V_{THN} and less than $V_{DD} - V_{THP}$ (make sure that this is understood). ■

Example 11.4

Show, using SPICE and the long-channel process, the transfer curves for the CMOS inverter with transconductance ratios β_n/β_p of 3, 1, and 1/3. Explain what changing the inverter ratio does to the transfer characteristics.

The simulation results are seen in Fig. 11.8. For all three DC sweeps the MOSFET's lengths are 1. For the case when $\beta_n/\beta_p = 1$, the $W_n = 10$ and $W_p = 30$. For the case when $\beta_n/\beta_p = 3$, the $W_n = 10$ and $W_p = 10$, etc. Increasing the strength of the NMOS (increasing the NMOS's width, which decreases R_n) causes the switching point voltage to decrease. We also get a decrease in V_{SP} by decreasing the strength of the PMOS device (which increases R_p). ■

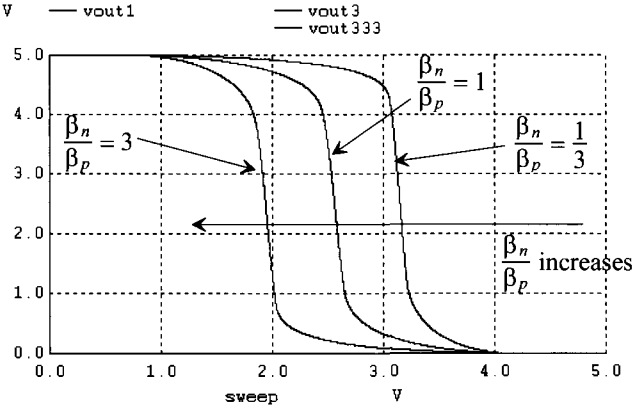


Figure 11.8 Sizing the inverter changes the switching point voltages.

11.2 Switching Characteristics

The switching behavior of the inverter can be generalized by examining the parasitic capacitances and resistances associated with the inverter. Consider the inverter shown in Fig. 11.9 with its equivalent digital model. Although the model is shown with both switches open, in practice one of the switches is closed, keeping the output connected to VDD or ground. The effective input capacitance of the inverter is

$$C_{in} = \frac{3}{2}(C_{ox1} + C_{ox2}) = C_{inn} + C_{inp} \quad (11.6)$$

The effective output capacitance of the inverter is simply

$$C_{out} = C_{ox1} + C_{ox2} = C_{outn} + C_{outp} \quad (11.7)$$

The intrinsic propagation delays of the inverter are

$$t_{PLH} = 0.7 \cdot R_{p2} \cdot C_{out} \text{ and } t_{PHL} = 0.7 \cdot R_{n1} \cdot C_{out} \quad (11.8)$$

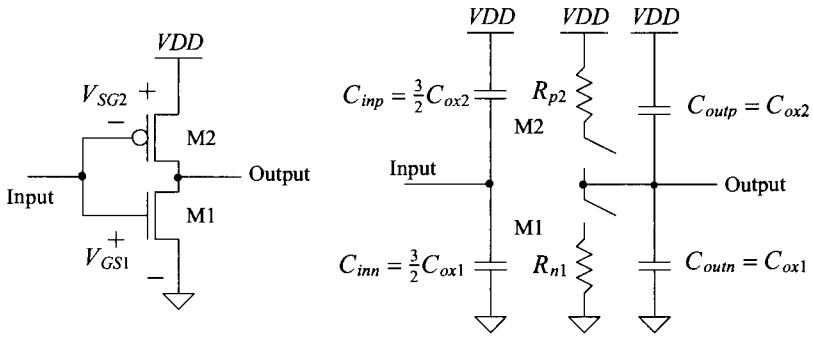


Figure 11.9 The CMOS inverter switching characteristics using the digital model.

Example 11.5

Estimate and simulate the intrinsic propagation delays of the inverter seen in Fig. 11.7. Estimate the inverter's input capacitance.

From the data in Table 10.2 and Eqs. (11.6) to (11.8), we can write

$$t_{PHL} = t_{PLH} = 0.7 \cdot 3.4k \cdot (0.625 + 1.25) \text{ fF} = 4.5 \text{ ps}$$

The simulation results are seen in Fig. 11.10. The intrinsic delays are considerably larger than this calculation (around 20 ps).

Using Eq. (11.6), the inverter's input capacitance is

$$C_{in} = \frac{3}{2}(0.625 + 1.25) \text{ fF} = 2.8 \text{ fF}$$

Sizing up the width of the PMOS so that its effective resistance is equal to R_n has the unwanted effect of increasing the inverter's input capacitance. ■

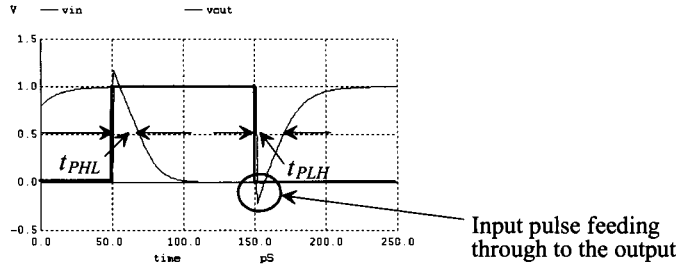


Figure 11.10 The intrinsic propagation delays of an inverter.

The propagation delays for an inverter driving a capacitive load are

$$t_{PLH} = 0.7 \cdot R_{p2} \cdot C_{tot} = 0.7 \cdot R_{p2} \cdot (C_{out} + C_{load}) \quad (11.9)$$

and

$$t_{PLH} = 0.7 \cdot R_{n1} \cdot C_{tot} = 0.7 \cdot R_{n1} \cdot (C_{out} + C_{load}) \quad (11.10)$$

where C_{tot} is the total capacitance on the output of the inverter, that is, the sum of the output capacitance of the inverter, any capacitance of interconnecting lines, and the input capacitance of the following gate(s).

Example 11.6

Estimate and simulate the propagation delays for the circuit seen in Fig. 11.11. Use the 50 nm CMOS process.

Because the load capacitance is much larger than the output capacitance of the inverter, we can rewrite

$$t_{PLH} = 0.7 \cdot R_{p2} \cdot C_{tot} \approx 0.7 \cdot R_{p2} \cdot C_{load} = 120 \text{ ps}$$

and

$$t_{PLH} = 0.7 \cdot R_{n1} \cdot C_{tot} \approx 0.7 \cdot R_{n1} \cdot C_{load} = 120 \text{ ps}$$

The simulation results are seen in Fig. 11.11. Notice that we are treating the MOSFETs used in the digital circuits as resistors and calculating the delays with a simple product of these resistors with the load capacitance. ■

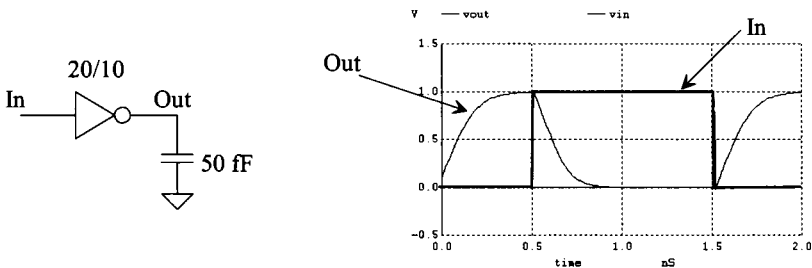


Figure 11.11 The delay associated with an inverter driving a 50 fF load.

The Ring Oscillator

The odd number of inverters in the circuit shown in Fig. 11.12 forms a closed loop with positive feedback and is called a ring oscillator. The oscillation frequency is given by

$$f_{osc} = \frac{1}{n \cdot (t_{PHL} + t_{PLH})} \quad (11.11)$$

assuming that the inverters are identical and n is the number (odd) of inverters in the ring oscillator. Since the ring oscillator is self-starting, it is often added to a test portion of a wafer to indicate the speed of a particular process run. The sum of the high-to-low and low-to-high delays is used to calculate the period of the oscillation because each inverter switches twice during a single oscillation period.

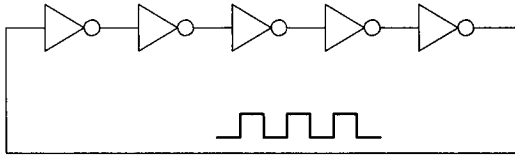


Figure 11.12 A five-stage ring oscillator.

When identical inverters are used the capacitance on the inverter's input/output is the sum of an inverter's input capacitance with the inverter's output capacitance, see Fig. 11.9, or

$$C_{tot} = \overbrace{C_{oxp} + C_{oxn}}^{C_{out}} + \overbrace{\frac{3}{2} \cdot (C_{oxp} + C_{oxn})}^{C_{in}} = \frac{5}{2} \cdot (C_{oxp} + C_{oxn}) \quad (11.12)$$

where, again, $C_{oxp} = C'_{ox} \cdot W_p \cdot L_p \cdot (scale)^2$ and $C_{oxn} = C'_{ox} \cdot W_n \cdot L_n \cdot (scale)^2$. The delay is then calculated using

$$t_{PHL} + t_{PLH} = 0.7 \cdot (R_n + R_p) \cdot C_{tot} \quad (11.13)$$

Dynamic Power Dissipation

Consider the CMOS inverter driving a capacitive load shown in Fig. 11.13. Each time the inverter changes states, it must either supply a charge to C_{tot} or sink the charge stored on C_{tot} to ground. If a square pulse is applied to the input of the inverter with a period T and frequency, f_{clk} , the average amount of current that the inverter must pull from VDD , recalling that current is being supplied from VDD only when the PMOS device is on, is

$$I_{avg} = \frac{Q_{C_{tot}}}{T} = \frac{VDD \cdot C_{tot}}{T} \quad (11.14)$$

The average dynamic power dissipated by the inverter is

$$P_{avg} = VDD \cdot I_{avg} = \frac{C_{tot} \cdot VDD^2}{T} = C_{tot} \cdot VDD^2 \cdot f_{clk} \quad (11.15)$$

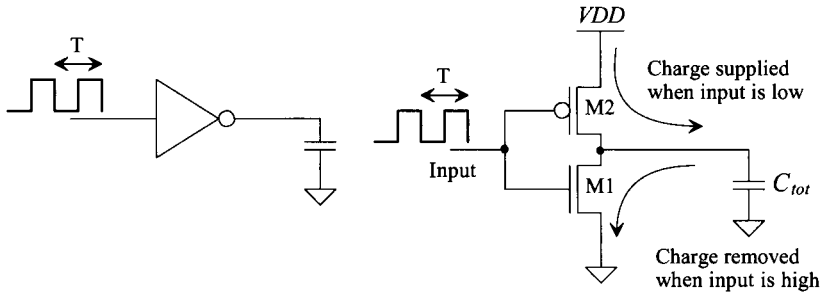


Figure 11.13 Dynamic power dissipation of the CMOS inverter.

Notice that the power dissipation is a function of the clock frequency, power supply voltage, and load capacitance. A great deal of effort is put into reducing the power dissipation in CMOS circuits. One of the major advantages of dynamic logic (Ch. 14) is its lower power dissipation.

To characterize the speed of a digital process, a term called the power delay product (*PDP*) is often used. The *PDP*, measured in Joules, is defined by

$$PDP = P_{avg} \cdot (t_{PHL} + t_{PLH}) \quad (11.16)$$

These terms can be determined from a ring oscillator. The *PDP* is frequently used to compare different technologies or device sizes; for example, a GaAs process can be compared with a 50 nm CMOS process. Although the GaAs process may have a lower propagation delay, the power dissipation may be larger and result in a larger *PDP*.

Example 11.7

Estimate the oscillation frequency of an 11-stage ring oscillator in the 50 nm process using the inverter seen in Fig. 11.7 (see Table 10.2). Verify the estimate with simulations.

Using the data in Table 10.2 and Eqs. (11.11) – (11.13), we can write

$$C_{tot} = \frac{5}{2} \cdot (1.25 + 0.625) \text{ fF} = 4.7 \text{ fF}$$

and

$$t_{PHL} + t_{PLH} = 0.7 \cdot (3.4k + 3.4k) \cdot 4.7 \text{ fF} = 22 \text{ ps}$$

For an 11-stage ring oscillator, we get an oscillation frequency of

$$f_{osc} = \frac{1}{11 \cdot (22 \text{ ps})} = 4.1 \text{ GHz}$$

The simulation results are seen in Fig. 11.14. The simulated oscillation frequency is close to 1.25 GHz or considerably different from our hand calculations (as will be the case when delays are close to the intrinsic values). The average power, P_{avg} , dissipated by a single inverter in this ring oscillator, using Eq. (11.15), is estimated as 19.6 μW . The *PDP* for the 50 nm process is then $431 \times 10^{-18} \text{ J}$. ■

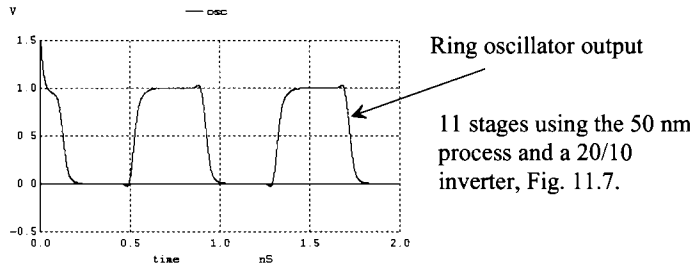


Figure 11.14 Oscillation frequency for the ring oscillator described in Ex. 11.7.

11.3 Layout of the Inverter

If care is not taken when laying out CMOS circuits, the parasitic devices present can cause a condition known as latch-up. Once latch-up occurs, the inverter output will not change with the input; that is, the output may be stuck in a logic state. To correct this problem, the power must be removed. Latch-up is especially troubling in output driver circuits.

Latch-Up

Figure 11.15 illustrates two methods of laying out an inverter. Notice how the cell's inputs and outputs are on metal2, while the power and ground conductors are routed on metal1 in the standard cell frame (see Fig. 4.15 and the associated discussion). The cross-sectional view in Fig. 11.16 shows both the NMOS and PMOS devices that make up an inverter (and associated parasitics). Notice that in Fig. 11.10, the input pulse feeds through the gate-drain capacitance of the MOSFETs to the output of the inverter. This causes the output to change in the same direction as the input before the inverter starts to switch. This feedthrough and the parasitic bipolar transistors can cause the latch-up.

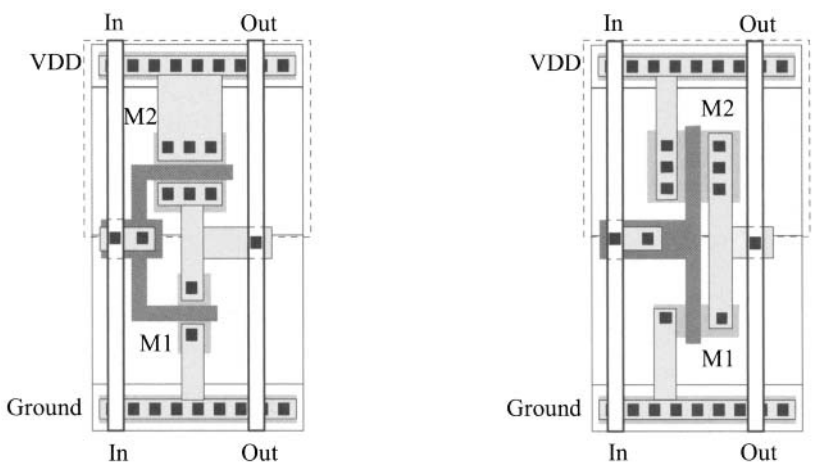


Figure 11.15 Layout styles for inverters.

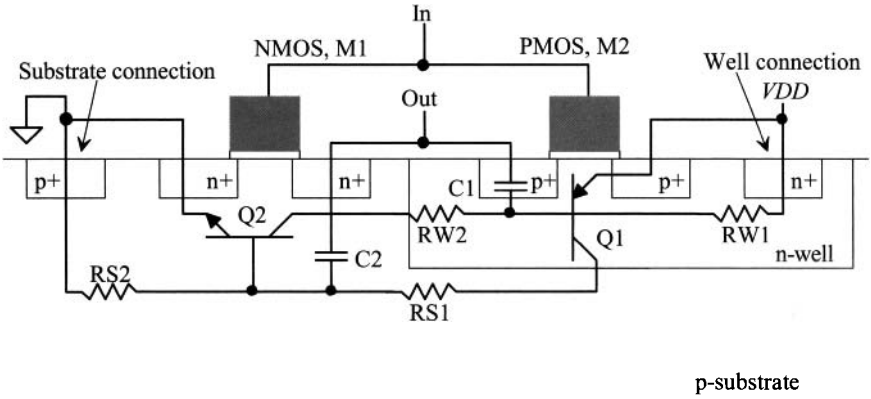


Figure 11.16 Cross-sectional view of an inverter showing parasitic bipolar transistors and resistors.

In Fig. 11.16, the emitter, base, and collector of transistor Q1 are the source of the PMOS, n-well, and substrate, respectively. Transistor Q2's collector, base, and emitter are the n-well, substrate, and source of the NMOS transistor. Resistors RW1 and RW2 represent the effects of the resistance of the n-well, and resistors RS1 and RS2 represent the resistance of the substrate. The capacitors C1 and C2 represent the drain implant depletion capacitance, that is, the capacitance between the drains of the transistors and the n-well (for C1) and substrate (for C2). The parasitic circuit resulting from the inverter layout is shown in Fig. 11.17.

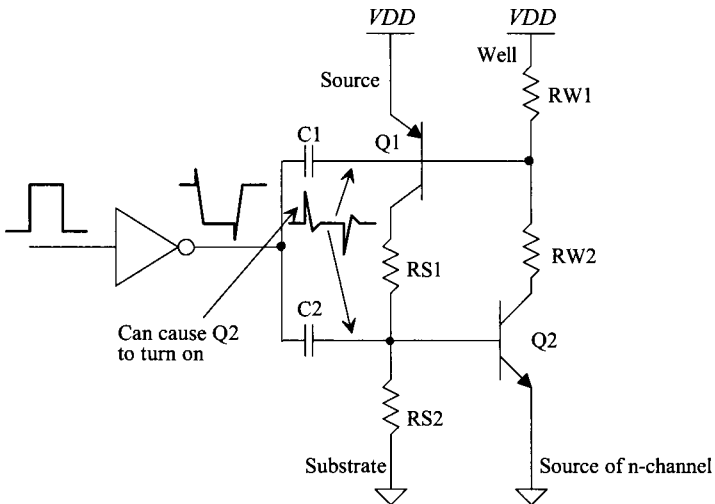


Figure 11.17 Schematic used to describe latch-up.

If the output of the inverter switches fast enough, the pulse fed through C2 (for positive-going inputs) can cause the base-emitter junction of Q2 to become forward biased. This then causes the current through RW2 and RW1 to increase, turning on Q1. When Q1 is turned on, the current through RS1 and RS2 increases, causing Q2 to turn on harder. This positive feedback will eventually cause Q2 and Q1 to turn on fully and remain that way until the power is removed and reapplied. A similar argument can be given for negative-going inputs feeding through C1, V_{DD} bouncing upwards, or ground bouncing downwards.

Several techniques reduce the latch-up problem. One technique is to slow the rise and fall times of the logic gates, reducing the amount of signal fed through C1 and C2. Reducing the areas of M1 and M2's drains lowers the size of the depletion capacitance and the amount of signal fed through. Probably the best method of reducing latch-up effects is to reduce the parasitic resistances RW1 and RS2. If these resistances are zero, Q1 and Q2 never turn on. The value of these resistances, as seen in Fig. 11.16, is a strong function of the distance between the well and substrate contacts. Simply put, the closer these contacts are to the MOSFETs used in the inverter, the less likely it is that the inverter will latch up. These contacts should be plentiful as well as close. Placing substrate and well contacts between the PMOS and NMOS devices provides a low-resistance connection to V_{DD} and ground, significantly helping to reduce latchup (see Fig. 11.18 for a simple layout example). Placing n+ and p+ areas between or around circuits reduces the amount of signal reaching a given circuit from another circuit. These implants are sometimes called guard rings (see Fig. 5.5). Notice that poly cannot be used to connect the gates of the MOSFETs, since poly over the n+ or p+ will be interpreted as a MOSFET. Therefore, metal2 is used to connect the MOSFETs together. The cost of reducing the possibility of latch-up is a more complicated layout in a larger area.

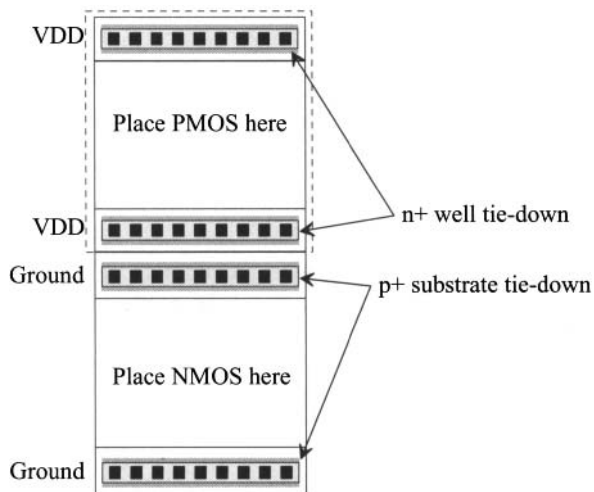


Figure 11.18 Adding an extra implant between NMOS and PMOS to reduce latch-up.

11.4 Sizing for Large Capacitive Loads

Designing a circuit to drive large capacitive loads with minimum delay is important when driving off-chip loads. As we saw in Ex. 11.6, a large load capacitance can drastically affect the delay through an inverter. Remembering our discussion in Sec. 10.3, we see that using a standard scope probe to measure an output signal can result in delays that are microseconds in length. In order to avoid this situation, we add a buffer circuit (a string of inverters) between the on-chip logic and the bonding pads. In this section we discuss how to design (select the widths of the MOSFETs) for low delays.

Buffer Topology

Consider the inverter string (a buffer) driving a load capacitance, labeled C_{load} and shown in Fig. 11.19. Moving towards the load in a cascade of the N inverters, each inverter larger than the previous by a factor A (that is, the width of each MOSFET is multiplied by A), a minimum delay can be obtained as long as A and N are picked correctly. Each inverter's input capacitance is larger than the previous inverter's input capacitance by a factor of A ,

$$C_{in2} = A \cdot C_{in1} \text{ and } C_{in3} = A \cdot C_{in2} = A^2 \cdot C_{in1}, \text{ etc.} \quad (11.17)$$

The effective switching resistances are also divided by a factor of A , resulting in the same delay for each stage of the buffer,

$$R_{n,p2} = \frac{R_{n,p1}}{A} \text{ and } R_{n,p3} = \frac{R_{n,p2}}{A} = \frac{R_{n,p1}}{A^2} \quad (11.18)$$

In other words, the effective switching resistance of the NMOS in the third inverter is $1/A^2$ smaller than the effective switching resistance of the NMOS in the first inverter.

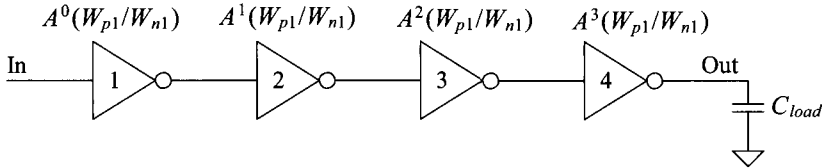


Figure 11.19 Cascade of inverters used to drive a large load capacitance.

If the load capacitance equals the input capacitance of the last inverter multiplied by A (so that the load capacitance has a value equal to the input capacitance of the next inverter if an additional inverter were used), then

$$\text{Input C of the final inverter} = C_{in1} \cdot A^N = C_{load} \quad (11.19)$$

or

$$A = \left[\frac{C_{load}}{C_{in1}} \right]^{\frac{1}{N}} \quad (11.20)$$

Again, the delays of each stage in the buffer are equal. The total delay of the inverter string is given by

$$(t_{PHL} + t_{PLH})_{total} = 0.7 \cdot \underbrace{(R_{n1} + R_{p1})(C_{out1} + AC_{in1})}_{\text{First-stage delay}} + 0.7 \cdot \underbrace{\frac{(R_{n1} + R_{p1})}{A} \cdot (AC_{out1} + A^2C_{in1})}_{\text{Second-stage delay}} \dots \quad (11.21)$$

Because as the inverters are increased in size by A , their capacitances, both input and output, increase by A , while their resistances decrease by a factor A . This equation can be rewritten as

$$(t_{PHL} + t_{PLH})_{total} = 0.7 \cdot \sum_{k=1}^N (R_{nk} + R_{pk})(C_{outk} + AC_{ink}) = 0.7 \cdot N(R_{n1} + R_{p1})(C_{out1} + AC_{in1}) \quad (11.22)$$

or with the help of Eq. (11.20):

$$(t_{PHL} + t_{PLH})_{total} = 0.7 \cdot N(R_{n1} + R_{p1}) \cdot \left(C_{out1} + \left(\frac{C_{load}}{C_{in1}} \right)^{\frac{1}{N}} \cdot C_{in1} \right) \quad (11.23)$$

The minimum delay can be found by taking the derivative of this equation with respect to N , setting the result equal to zero, and solving for N . Taking the derivative of Eq. (11.23) with respect to N gives

$$0.7 \cdot \left((R_{n1} + R_{p1})C_{out1} + (R_{n1} + R_{p1})C_{in1} \left(\left(\frac{C_{load}}{C_{in1}} \right)^{\frac{1}{N}} + N \cdot \left(\frac{C_{load}}{C_{in1}} \right)^{\frac{1}{N}} \frac{\ln(C_{load}/C_{in1})}{-N^2} \right) \right) = 0 \quad (11.24)$$

The first term in this equation is the intrinsic delay of the first inverter in our cascade of inverters. *This first stage represents, generally, the on-chip logic gate and is **not part of the buffer**.* The first inverter is included in the calculations to represent the limited drive of the on-chip logic. If we assume that this delay is small, solving for N gives

$$N = \ln \frac{C_{load}}{C_{in1}} \quad (11.25)$$

Equations (11.25) and (11.20) are used in the buffer design to drive a large capacitance. Note that the larger the first inverter, the fewer the number of inverters needed to drive a given capacitive load.

Example 11.8

Estimate $t_{PHL} + t_{PLH}$ for the inverter shown in Fig. 11.11 driving a load capacitance of 20 pF. Design a buffer to drive the load capacitance with a minimum delay.

The total propagation delay of the unbuffered inverter, Fig. 11.11, is given by

$$t_{PHL} + t_{PLH} = 0.7 \cdot (3.4k + 3.4k) \cdot (20 \text{ pF}) = 95 \text{ ns} !$$

Designing a buffer begins with determining C_{in1} . For our 20/10 inverter in the 50 nm process, the C_{in1} (see Table 10.2) is $\frac{2}{2}(1.25 + 0.625) \text{ fF} = 2.81 \text{ fF}$ and C_{out1} is 1.875 fF . To determine the number of inverters, we use

$$N = \ln \left(\frac{20 \text{ pF}}{2.81 \text{ fF}} \right) = 8.87 \rightarrow 9 \text{ stages}$$